INF 1003 Mathematics 1

Tutorial 3

1a) $\binom{9}{4} \times 5! = \frac{9\times8\times7\times6}{4\times3\times2\times1} \times 120 = 126 \times 120$

$= 15,120$ //

b) $\binom{8}{3} \times 5! = \frac{8\times7\times6}{3\times2\times1} \times 120 = 56 \times 120$

$= 6720$ //

c) $2 \times \binom{8}{4} \times 5! = 2 \times \frac{8\times7\times6\times5}{4\times3\times2\times1} \times 120$

$= 2 \times 70 \times 120$

$= 16,800$ //

2a) 8 remaining bits in the middle

$2^9 = 256$ //

b) $\binom{10}{4} = \frac{10\times9\times8\times7}{4\times3\times2\times1} = 210$ //

c) $\binom{10}{0} = 1$, $\binom{10}{1} = 10$, $\binom{10}{2} = 45$, $\binom{10}{3} = 120$, $\binom{10}{4} = 210$

$1 + 10 + 45 + 120 + 210 = 386$ //

d) $2^{10} - \left(\binom{10}{0} + \binom{10}{1} + \binom{10}{2} + \binom{10}{3}\right) = 1024 - (1 + 10 + 45 + 120)$

$= 1024 - 176$

$= 848$ //

3a) (BCD, A, E, F, G)

$5! = 120$

c) (BA, GF, C, D, E)

$5! = 120$

e) C is part of both

0 //

b) (CEFA, B, D, G)

$4! = 24$

d) (ABC, DE, F, G)

$4! = 24$

f) (CBA, BED, F, G)

$3! = 6$

4) $8! \times \binom{9}{5} \times 5! = 40{,}320 \times 126 \times 120$
$= 609{,}813{,}120$ //

5) $\binom{10}{3} \times \binom{15}{3} = 120 \times 455$
$= 54{,}600$ //

6) $101 \times 4 + 1 = 405$ //

7a) (1,8), (2,7), (3,6), (4,5)
there are 4 pairs that sum to 9. by the pigeonhole principle, if we choose 5 numbers from these 8, at least one pair will be selected.
the pigeons are the 5 selected numbers, the pigeonholes are the four pairs.

b) if we choose 4 numbers, it is possible not to select any pairs that sum to 9. for example, selecting {1,2,3,4} avoids all the pairs.
thus the conclusion is not true for 4 integers.

8) pigeons → 51 houses, pigeonholes → 100 addresses
by the pigeonhole principle, at least two houses must have consecutive addresses.